TECHNICAL APPROACH REPORT

Cynbit LeadIQ

AI-Powered Inquiry Intelligence & Lead Routing System

Prepared for: Cynbit Technologies
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Project: AI/ML Proof of Concept | **Environment:** Google Colab | **Language:** Python

Scope: Proposed processing layer behind existing inquiry channels. The prototype uses synthetic data and does not connect to Cynbit's live forms, CRM, email, or customer database.

1. Introduction

Cynbit LeadIQ is a proof of concept for processing inquiries from Blueprint, service inquiries, product demo requests, and general contact. It classifies inquiry text, extracts available details, assigns an explainable priority, recommends a team and next action, and drafts a follow-up message for human review.

2. System Architecture

Workflow: Existing inquiry channels → data collection and standardization → NLP classification → information extraction → priority scoring → team recommendation → dashboard / CSV output → human review. The prototype supports batch CSV processing and individual inquiry analysis through Gradio. Both modes use the same central analysis function.

3. Technology Stack

Technology	Purpose
Python / Google Colab	Core development and execution
Pandas / NumPy	Dataset handling and processing
Sentence Transformers: all-MiniLM-L6-v2	Sentence embeddings for semantic classification
Scikit-learn	Cosine similarity and TF-IDF fallback
Python Regex	Budget and timeline extraction
Rule-based Python logic	Priority scoring and team routing
Gradio / Matplotlib	Interactive demo and visualizations
CSV	Input and output storage

4. Model Selection and Classification

The pretrained Sentence Transformers model **all-MiniLM-L6-v2** converts inquiry text and predefined service/product descriptions into embeddings. Cosine similarity compares the inquiry with each category description; the highest-scoring category is selected. Categories include AI Automation & Agents, Custom Software Development, Cloud & DevOps, CRM & Sales Automation, Digital Marketing & SEO, Cynbit Studio, TenantEase, Revio, and General / Needs Discovery.

If the embedding model cannot initialize, TF-IDF vectorization with cosine similarity is used as a fallback. The model is pretrained, not fine-tuned on Cynbit-specific data. Similarity is semantic closeness, not prediction probability or accuracy.

5. Data Pipeline and Information Extraction

The synthetic **sample_inquiries.csv** contains 16 records: inquiry ID, source channel, inquiry text, provided category, industry, budget, timeline, and product. Pandas loads and prepares each record. Python Regex extracts budget and timeline when structured values are absent (for example, 'Rs 5 lakh' → ■500,000; 'start next week' → 'next week'). Structured

budget/timeline inputs take precedence. Industry and source channel are supplied fields, not automatically predicted.

The central function **analyze_inquiry()** performs classification, extraction, scoring, routing, and follow-up draft generation, returning structured results.

6. Priority Scoring and Routing

Scoring signal	Points
Budget provided	+20
Urgent timing	+25
Non-urgent timeline provided	+15
Action-oriented intent	+20
3+ requirement keywords	+25
1–2 requirement keywords	+15
No requirement keywords	+5
Product demo request	+10

Timeline scores and requirement-detail scores are alternatives. The total is capped at 100. Priority levels: High (70–100), Medium (40–69), Low (0–39). Scoring reasons are recorded for transparency. This is heuristic business logic, not a trained lead-conversion model. Team routing uses predefined mappings from category to recommended team and next action; these should be validated against Cynbit's actual team structure.

7. Prompt Engineering and Response Generation

The current prototype does not use an external LLM API or generative prompt. It combines semantic NLP classification with Python extraction, scoring, routing, and template-based follow-up drafts. Drafts require human review and are not sent automatically. A future LLM could return structured summaries, requested service, requirements, budget, timeline, missing information, and next action, with explicit instructions not to invent missing details.

8. Execution Plan

Phase	Activity
1	Understand inquiry workflow and define problem
2	Prepare synthetic inquiry dataset
3	Implement NLP classification and fallback
4	Develop budget and timeline extraction
5	Implement priority scoring and routing
6	Build Gradio interface
7	Implement batch processing and CSV export
8	Test scenarios and document limitations

Execution: Open the notebook in Colab, install dependencies, upload the sample CSV, run cells sequentially, test Gradio, and run the export cell to download the analyzed CSV.

9. Testing and Evaluation

Test Blueprint, service, product demo, and general inquiries with varied budgets, timelines, and urgency. A future labeled test set can be evaluated using accuracy, precision, recall, and F1-score. Extraction can be checked against manually verified values. No independent labeled test set or accuracy claims are included in this prototype.

10. Limitations and Future Enhancements

Limitations: synthetic data; pretrained, non-fine-tuned model; source channel and industry supplied as inputs; basic Regex extraction; rule-based priority and routing; no live website, CRM, email, or database integration; and Gradio submissions are not automatically appended to the batch CSV.

Future work: secure API/webhook integration; LLM-based requirement extraction; predictive lead scoring using sufficient historical outcomes; CRM integration after workflow discovery; duplicate detection; centralized analytics; and feedback-based improvement with human approval.

11. Conclusion

Cynbit LeadIQ demonstrates how semantic NLP and explainable business rules can support inquiry processing. It combines Sentence Transformers classification, Regex extraction, heuristic priority scoring, predefined routing, batch CSV processing, and an interactive Gradio demo. It is a synthetic-data proof of concept and proposed enhancement layer, not a live Cynbit integration.

12. References

1. Cynbit Official Website
2. Cynbit Contact / Blueprint
3. Cynbit Products
4. Cynbit Services
5. Sentence Transformers
6. all-MiniLM-L6-v2
7. Scikit-learn
8. Gradio